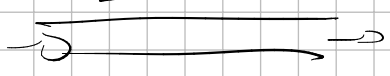
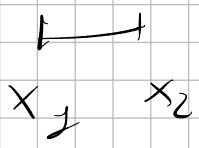
$$\rho(t,x) = \text{"}\frac{\#\ \text{particelle}}{\text{volume}}\text{"} = \frac{\text{massa del gas}}{\text{volume del gas}}$$

1D

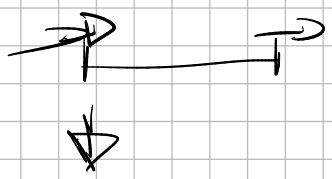




$$\int_{x_1}^{x_2} \rho(t,x)\,dx$$ = MASSA DEL NOSTRO GAS in $[x_1,x_2]$ AL VARIARE DEL TEMPO

$v(t,x)$ velocità del gas

MASSA SI CONSERVA A MENO DI EFFETTI AL BORDO



FLUSSO

$$\underbrace{v(t,x_2)\cdot\rho(t,x_2)}$$

NEL TEMPO $[t_1,t_2]$

VARIAZIONE IN TEMPO DI MASSA

$$\int_{x_1}^{x_2} \rho(t_2,x) - \rho(t_1,x)\,dx =$$

$$= -\int_{t_1}^{t_2} \rho(t,x_2)\,v(t,x_2) - \rho(t,x_1)\,v(t,x_1)\,dt$$

QUANTA MASSA È ENTRATA / USCITA CON SEGNO AI BORDI

$$\frac{d}{dt}\int_{x_1}^{x_2} \rho(t,x)\,dx = -\rho(t,x_2)\cdot v(x_2,t) + \rho(t,x_1)\,v(x_1)$$

1) $\frac{d}{dt}\rho(t,x) = -\frac{\partial}{\partial x}\underbrace{\left(\rho(t,x)\, v(t,x)\right)}_{f(\rho(t,x))}$?

CONSERVAZ. MASSA

$$\boxed{\frac{d}{dt}\rho(t,x) + \partial_x f(\rho(x,t)) = 0}$$

CONSERVAZIONE MOMENTO

2) $\partial_t(\rho v) + \partial_x(\rho v^2 + p) = 0$

ORA SAPPIAMO LA VELOCITÀ

?

$p(\rho)$

CONSERVAZIONE DELL'ENERGIA

3) $\partial_t(\rho E) + \partial_x(\rho v E + v \cdot p) = 0$

$p \Leftrightarrow E$ EQUAZIONE DI STATO

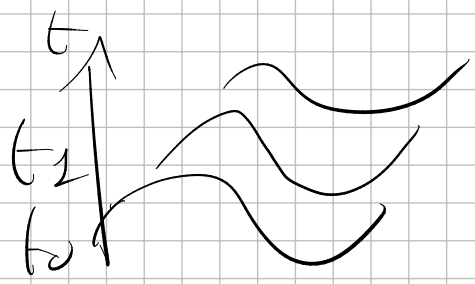
$p = (\gamma - 1)\cdot\rho\left(E - \frac{v^2}{2}\right)$ γ COSTANTE ADIABATICA

$E = \frac{p}{\rho(\gamma-1)} + \frac{v^2}{2}$ $\gamma = \frac{7}{5}$

TRASPORTO LINEARE

$\partial_t u(t,x) + a\,\partial_x u(t,x) = 0$ $a \in \mathbb{R}$

$u(t,x) = u_0(x - at)$



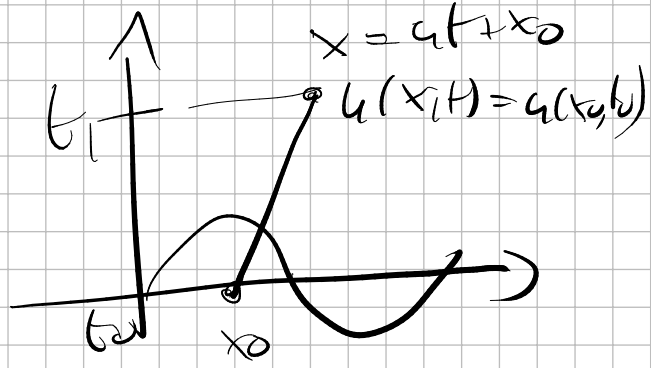
• CARATTERISTICHE

$$x = at + x_0 = x(t)$$

$$0 = \frac{d\,u(t, x(t))}{dt} = \partial_t u(t, x(t)) + \partial_x u(t, x(t)) \cdot \underbrace{\partial_t x(t)}$$

$$\Rightarrow \quad \partial_t x(t) = -\frac{\partial_t u}{\partial_x u} = a \quad = \text{costante}$$

$$x(t) = x_0 + at$$



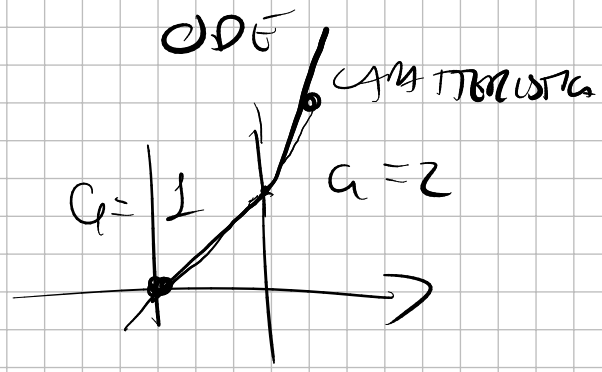
$$\Downarrow$$

$$u_t + a(x) \cdot u_x = 0$$

$$\partial_t x(t) = a(x(t)) \Rightarrow \text{RISOLVERE ODE}$$

$\Downarrow$

CARATTERISTICA



NON È UNA LEGGE DI CONSERVAZIONE

$$\int_{x_1}^{x_2} \partial_t u + \underbrace{\int_{x_1}^{x_2} a(x) \cdot u_x} = 0$$

$\Downarrow$

A PRIORI NON È UNA DIFFERENZA DI FLUSSI

$$u_t + (a(x)\cdot u)_x = 0$$

LO È UNA LEGGE DI CONSERVAZIONE!

$$\underbrace{u_t + a(x)\cdot u_x}_{\text{cerchiamo caratteristiche di questa parte}} + \underbrace{a_x(x)\cdot u(x)}_{\text{non facciamo ODE}} = 0$$

$$\partial_t X(t) = a(x) \quad \forall x_0 \longrightarrow \text{ODE \#1}$$

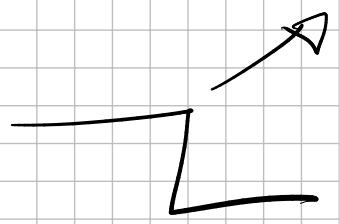
CARATTERISTICHE

$$\frac{d\,u(t,x(t))}{dt} = -a_x(x(t))\cdot u(x(t))$$

ODE #2

$u \in L^2(\mathbb{R}^+ \times \Omega)$

$[x_1, x_2] = \Omega \subseteq \mathbb{R}$



$$u_t + \partial_x(a(x)\cdot u(t,x)) = 0 \qquad \varphi \in C^\infty(\mathbb{R}^+ \times \Omega)$$

$$\int_0^T \int_\Omega \varphi(x,t)\cdot u_t(x,t) + \varphi(x,t)\,(a(x)\cdot u(t,x))_x = 0$$

INTEGR. PER PARTI IN t

I.P.P.

$$\int_\Omega \varphi(x,T)\, u(x,T) - \varphi(x,0)\, u(x,0)\, dx$$

$$- \int_0^T \int_\Omega \partial_t \varphi(x,t) \cdot u(x,t)\, dt\, dx + \int_0^T \varphi(x_R,t)\, a(x_R)\, u(x_R,t)\, dt$$

$$- \int_0^T \varphi(x_L,t)\, a(x_L)\, u(x_L,t)\, dt$$

$$- \int_0^T \int_\Omega \partial_x \varphi(x,t) \cdot a(x)\, u(x,t)\, dx\, dt = 0$$

HA DEFINITAMENTE SENSO PER $u \in L^2$

SOL DI PDE DEBOLE È TROVA $a \in L^2$:

$$\forall \varphi \in C^\infty$$

Se scelgo $\varphi = u$ $\qquad \frac{u^2}{2}$ = "ENERGIA"

$$\int_{t_0}^{t_1} \int_{x_L}^{x_R} u \cdot \partial_t u + \int_{t_0}^{t_1} \int_{x_L}^{x_R} u\, \partial_x (a \cdot u(x,t))\, dx\, dt = 0$$

$$\int_{t_0}^{t_1} \underbrace{\int_{x_L}^{x_R} \partial_t \left(\frac{u^2}{2}\right)} + \int_{t_0}^{t_1} \int_{x_L}^{x_R} \partial_x \left(a \frac{u^2}{2}\right) dx\, dt = 0$$

$$\int_{x_L}^{x_R} \frac{u^2(t_1)}{2} - \frac{u^2(t_0)}{2}\, dx + \int_{t_0}^{t_1} a \frac{u^2(x_R)}{2} - a \frac{u(x_L)^2}{2}\, dt = 0$$

CONDIZ. BORDO ONO DIRI / PERIODICHE

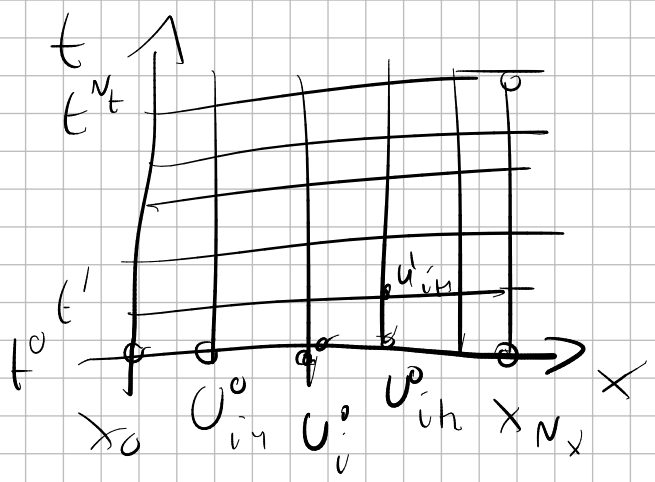
$$\int_{x_L}^{x_R} \frac{u^2(t_1)}{2} = \int_{x_L}^{x_R} \frac{u^2(t_0)}{2}$$

ENERGIA SI CONSERVA (A MENO DI CONTRIBUTI DAI BORDI)

$$\partial_t u^\varepsilon + a\,\underbrace{\partial_x u^\varepsilon} = \underbrace{\varepsilon\, u^\varepsilon_{xx}}_{\text{VISCOSITÀ}}$$

$$\lim_{\varepsilon \to 0} u^\varepsilon(t,x) = \underbrace{u(t,x)}_{\text{SOLUZIONI A VISCOSITÀ EVANESCENTE}}$$

$$\partial_t u + \partial_x u = 0 \qquad a = 1$$



$$\underbrace{(\partial_x u)_i \approx \frac{u_{i+1} - u_{i-1}}{2\Delta x}}$$

$$\approx \frac{1}{\Delta x}$$

$$u_i^n \approx u(t^n, x_i)$$

EULERO ESPLICITO (STABILE) $\Delta t \sim \Delta x$

$$\frac{u_j^{n+1} - u_j^n}{\Delta t} + \frac{u_{j+1}^n - u_{j-1}^n}{2\Delta x} = 0$$

È STABILE? STABILITÀ DI VON NEUMANN

$$u_j^n = e^{i \underbrace{j\Delta x}_{x} \cdot k} \cdot c_k^n \quad k \in \mathbb{Z}$$

$$c_k^{n+1} \cdot \underbrace{e^{i j \Delta x k}} = c_k^n \underbrace{e^{i j \Delta x k}} - \frac{\Delta t}{2\Delta x} \cdot$$

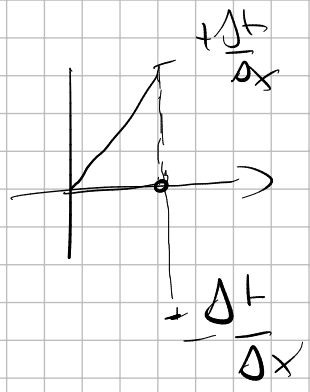
$$\cdot \left(e^{i (j+1) \Delta x k} - e^{i (j-1) \Delta x k} \right) c_k^n$$

$$c_k^{n+1} = c_k^n \underbrace{\left(1 - i \frac{\Delta t}{\Delta x} \boxed{\left(\frac{e^{i \Delta x k} - e^{-i \Delta x k}}{2i} \right)} \right)}_{g(k)}$$

COEFFICIENTE AMPLIFICAZ.

$$|g(k)| \leq 1 + \alpha \Delta t \qquad \alpha \text{ INDIP. DA } \Delta t$$

$$g(k) = 1 - i \frac{\Delta t}{\Delta x} \sin(k \Delta x)$$



$$1 \leq |g(k)|^2 \leq 1 + \frac{\Delta t^2}{\Delta x^2}$$

$$\Delta t \simeq \Delta x^2$$

$$\frac{u_j^{n+1} - u_j^n}{\Delta t} + \frac{u_{j+1}^{n+1} - u_{j-1}^{n+1}}{2\Delta x} = 0$$

$$c_k^{n+1}\left(\frac{1}{\Delta t} + \frac{1}{\Delta x}\underbrace{\left(\frac{e^{ik\Delta x} - e^{-ik\Delta x}}{2}\right)}_{i\sin(k\Delta x)}\right) = c_k^n \frac{1}{\Delta t}$$

$$c_k^{n+1} = \frac{1}{1 + \frac{\Delta t}{\Delta x} i\sin(k\Delta x)} c_k^n$$

$$|g(k)| = \underbrace{\left|1 + i\frac{\Delta t}{\Delta x}\sin(k\Delta x)\right|}_{\geq 1}^{-1} \leq 1$$

LAX-FRIEDRICHS

$$u_j^{n+1} = \frac{u_{j+1}^n + u_{j-1}^n}{2} - a\frac{\Delta t}{2\Delta x}(u_{j+1}^n - u_{j-1}^n)$$

$$= u_j^n + \underbrace{\frac{u_{j+1}^n - 2u_j^n + u_{j-1}^n}{2}}_{\simeq \Delta x^2 u_{xx} \simeq \varepsilon \cdot u_{xx}} - a\frac{\Delta t}{2\Delta x}(u_{j+1}^n - u_{j-1}^n)$$

$$g(k) = 1 + \underbrace{\frac{e^{ik\Delta x} - 2 + e^{-ik\Delta x}}{2}} - ia\frac{\Delta t}{\Delta x}\sin(k\Delta x)$$

$$= \cancel{1} + \cos(k\Delta x) - \cancel{1} - i\frac{a\Delta t}{\Delta x}\sin(k\Delta x)$$

$$= \cos(k\Delta x) - i\frac{a\Delta t}{\Delta x}\sin(k\Delta x)$$

$$|g(k)|^2 = \cos(k\Delta x)^2 + \left(\frac{a\Delta t}{\Delta x}\right)^2 \sin(k\Delta x)^2$$

se $\left(\frac{a\Delta t}{\Delta x}\right)^2 \leq 1$

$\left(|a+ib| = a^2+b^2\right)$

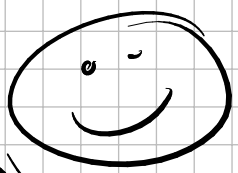
$\Downarrow$

$|g(k)|^2 \leq \cos^2 + \sin^2 = 1$

$$\left|a\frac{\Delta t}{\Delta x}\right| \leq 1 \qquad \Delta t \leq \frac{\Delta x}{|a|}$$

$\Downarrow$

STABILITÀ PER $\Delta t \sim \Delta x$



$$\Delta t \leq \min_x\left(\frac{\Delta x}{|a(x)|}\right) = \frac{\Delta x}{\max|a(x)|}$$

$$\Delta t \leq \frac{\Delta x}{|a|} \Rightarrow CFL$$

$$\Delta t = CFL \cdot \frac{\Delta x}{\max_x |a(x)|}$$

$CFL \leq 1$

$CFL \approx 0.9$